YIMIN ZHAO

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SUMMARY

- PhD candidate in Biostatistics at the University of Washington working on trustworthy medical AI: evaluation of LLM and agentic systems for factuality, evidence grounding, and knowledge currency. Lead PanCanBench and contribute to Agents' Last Exam and Terminal-Bench Science.
- Interned at Microsoft on Copilot Studio, improving an enterprise agentic retrieval-augmented generation (Agentic RAG) system with a new Copilot SDK retrieval tool and a self-evolving agent framework.
- Led the development of an end-to-end rubric-based LLM evaluation system covering 282 real patient questions and 3,639 clinician-written rubric criteria, benchmarking 22 leading LLMs; the work is under review at NEJM AI.
- 5+ years of experience modeling high-dimensional biological sequencing data, with results published in leading international journals including Nature Communications and Bioinformatics.

SKILLS

ML/AI	LLM evaluation (Rubrics, Factuality), Self-evolving Agents (Skills, Harness), Reinforcement Learning, RAG, Prompt Engineering, Deep Learning, Generative Models, Conformal Inference, Semi-supervised Learning, A/B Testing
Software	Python (PyTorch, NumPy, Pandas, LangChain), R, Shell, SQL

EDUCATION

University of Washington, School of Public Health — Seattle, WA
Ph.D. in Biostatistics (*Advisor: Jeff Leek, Mike Wu, Kevin Lin*) — Sep. 2022 - May 2027

Johns Hopkins University, Bloomberg School of Public Health — Baltimore, MD
ScM in Biostatistics (*Advisor: Hongkai Ji*) — Sep. 2020 - May 2022

Wuhan University, School of Mathematics and Statistics — Wuhan, China
B.S. in Mathematics and Applied Mathematics — Sep. 2015 - Jun. 2019

WORK EXPERIENCE

Microsoft — Jun. 2026 – Sep. 2026
PhD Intern, Applied Science and Research Team, Copilot Studio — Redmond, USA

- Developed a self-evolving agent framework that optimizes agent skills and proposes new tools, improving the success rate of an enterprise agentic RAG system grounded in SharePoint content by 5.5 percentage points (81.2% → 86.7%).
- Designed `file_evidence`, a Copilot SDK tool that reduced simple file-retrieval latency by 80%.

PUBLICATIONS AND RESEARCH

[1] **Yimin Zhao**, Sheela R Damle, Simone E Dekker, Scott Geng, Karly Williams Silva, Jesse J Hubbard, Manuel F Fernandez, Fatima Zelada-Arenas, Alejandra Alvarez, Brianne Flores, Alexis Rodriguez, Stephen Salerno, Carrie Wright, Zihao Wang, Pang Wei Koh, Jeffrey T Leek (2026). PanCanBench: A Comprehensive Benchmark for Evaluating Large Language Models in Pancreatic Oncology. *Under review at NEJM AI.* 🤗 Dataset on Hugging Face.

- Developed an end-to-end pipeline to collect, standardize, and polish expert-rubric annotations from medical fellows for evaluating LLM responses to patient-focused pancreatic cancer questions.
- Curated a dataset of 282 real-world pancreatic cancer patient questions with 3,639 fine-grained, clinician-authored rubrics; systematically evaluated the response quality of 22 LLMs.
- Introduced factuality verification beyond rubric scoring and found that some models with high rubric scores still made substantial factual errors, revealing the limitations of using rubric scores as the only metric for LLM evaluation.

[2] **Yimin Zhao**, Connor Finkbeiner, Manu Setty, Kevin Z. Lin (2026). SCOPE: Localizing fate-decision states and their regulatory drivers in single-cell differentiation. *Under major revision at Nature Communications.*

- Developed SCOPE, a machine learning framework for localizing cell-fate decision points from single-cell sequencing data using semi-supervised learning, graph-based label propagation, and conformal inference.
- Validated the framework on lineage-tracing and multi-omics datasets containing over 100,000 cells, revealing branchpoint-associated regulatory programs during cellular differentiation.

[3] Yiyou Sun, Xinyang Han, Weichen Zhang, Yuanbo Pang, Tianyu Wang, Yuhan Cao, Yixiao Huang, Chris Duroiu, Haoyun Zhang, Jeffrey Lin, **Yimin Zhao**, et al. (2026). Agents' Last Exam. *NeurIPS 2026*.

- Contributed to building Agents' Last Exam (ALE), a benchmark that evaluates agents on long-horizon, economically valuable real-world tasks through verifiable task outcomes.

[4] Zhen Zeng, Tianbei Zhang, Jiajia Zhang, Shuai Li, Sydney Connor, Boyang Zhang, **Yimin Zhao**, Jordan Wilson, Dipika Singh, Rima Kulikauskas, Candice D. Church, Thomas H. Pulliam, Saumya Jani, Paul Nghiem, Suzanne L. Topalian, Patrick M. Forde, Drew M. Pardoll, Hongkai Ji, Kellie N. Smith (2025). A minimal gene set characterizes multiple classes of tumor-specific TIL among different cancer types. *Nature Communications*, 16(1): 1070.

[5] Zhanying Feng, Xianwen Ren, Yuan Fang, Yining Yin, Chutian Huang, **Yimin Zhao** and Yong Wang (2020). scTIM: Seeking Cell-Type-Indicative Marker from single cell RNA-seq data by optimization model. *Bioinformatics*, 36(8), 2474-2485.

Ongoing Projects

Benchmarking LLMs on Evolving Medical Knowledge Feb 2026 – Present
Supervisor: Jeff Leek, Fred Hutch Seattle, USA

- Developed a dynamic benchmark for evaluating whether LLMs can resolve conflicts between outdated medical knowledge and newly released authoritative evidence.
- Curated evolving and matched static control facts from FDA drug labels, Phase III clinical trials, and clinical guidelines.
- Benchmarked state-of-the-art LLMs under closed-book, web-search, and oracle-evidence settings to quantify failures in knowledge updating, retrieval, and evidence integration.

Beyond Exchangeability: A Comprehensive Benchmark of Conformal Inference Under Distribution Shifts
Paper to be submitted Mar 2026 – Present
Supervisor: Kevin Lin, University of Washington; Mike Wu, Fred Hutch Seattle, USA

- Built a systematic benchmark of 8 conformal prediction methods for covariate shift, spanning propensity-score-based, distributionally robust, conditional-guarantee, and covariate-balancing approaches.
- Adapted entropy balancing from causal inference to conformal prediction, and showed it is more robust than propensity-score-based methods, whose performance depends on the propensity-score model and the data dimension.
- Showed that distributionally robust methods hinge on the divergence radius, which trades coverage against interval width.

Multi-omics Data Analysis Recommendation with RAG Apr 2025 – Present
Supervisor: Mike Wu, Fred Hutch Seattle, USA

- Developed a RAG-based tool for multi-omics analysis recommendation that mitigates hallucinations and delivers concrete, citation-backed guidelines compared with state-of-the-art LLM baselines.